

DBMS — Lecture 4

Object-Oriented Data Model

Learning Objectives

Understand the object-oriented data model.

Identify objects, classes, methods, and inheritance.

Compare object-oriented and relational approaches.

Object-Oriented Data Model

The **object-oriented data model** stores data as objects that combine data and behavior.

It is useful for applications with complex data such as multimedia, engineering, scientific, and CAD systems.

Core Concepts

Object: Entity containing state (data) and behavior (methods).

Class: Blueprint describing common properties and behavior of objects.

Encapsulation: Combines data and methods and controls access to internal details.

Inheritance: A subclass can inherit properties and behavior from a superclass.

Object Identity: Each object has a unique identity independent of its attribute values.

Example

Class: **Employee**

Attributes: Emp_ID, Name, Salary

Methods: calculateSalary(), displayDetails()

Subclass: **Manager** can inherit Employee features.

Relational vs Object-Oriented

Relational	Object-Oriented
Tables, rows, columns	Objects and classes
Relationships through keys	Object references/relationships
SQL commonly used	Object-oriented interfaces/languages commonly used
Strong for structured tabular data	Strong for complex objects and behavior

Key Takeaway

The **object-oriented model** combines data and behavior and supports concepts such as classes, inheritance, and encapsulation.